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COLLEGE OF AGRICULTURE, GKVK

BANGALORE – 560065

EAI421 – APPLICATION OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PROJECT TITLE

Paddy Yield Prediction Using Machine Learning: A Comparative Study of Linear Regression, Decision Tree, and Random Forest Models

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Abstract

Paddy (Oryza sativa) is one of the most cultivated staple crops across Asia, and accurate prediction of its yield is central to food security planning and resource management. This project investigates the application of supervised Machine Learning (ML) algorithms to predict paddy yield based on three climate-agronomic features: minimum temperature, maximum temperature, and relative humidity, all recorded during days 91–120 of the crop growth period.

A cleaned dataset of 74 observations sourced from field trials was used. Three regression models — Linear Regression, Decision Tree Regressor, and Random Forest Regressor — were trained and evaluated using Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 Score. Hyperparameter tuning via GridSearchCV was applied to all models. Results indicate that the Decision Tree Regressor outperformed other models with the lowest MSE (2.32×10^6) and MAE (1255.09 Kg), emerging as the best model. Feature importance analysis revealed that Minimum Temperature contributes most significantly (~76%) to the prediction, followed by Maximum Temperature (~22%).

Despite the models exhibiting negative R^2 scores — indicating high variance in the small dataset — this study establishes a baseline ML framework for climate-driven paddy yield forecasting. Future work should incorporate expanded datasets, additional agronomic features such as soil properties and irrigation levels, and deep learning approaches to improve predictive accuracy.

Keywords: *Paddy Yield Prediction, Machine Learning, Decision Tree, Random Forest, Linear Regression, Ridge Regression, Hyperparameter Tuning, Feature Importance, Precision Agriculture, Climate Variables*

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1. Introduction

Agriculture is the backbone of economic development in many developing nations, and crop yield prediction forms a cornerstone of modern food security strategy. Paddy (Oryza sativa), commonly known as rice, is the most widely cultivated cereal crop across Asia and feeds more than half the world's population. In India alone, paddy is cultivated across approximately 43 million hectares annually, and yield variability driven by climate change, weather patterns, and agronomic management presents significant challenges to farmers, policymakers, and food supply chains.

Traditional methods of yield estimation — based on field surveys, historical averages, and expert agronomic knowledge — are time-consuming, resource-intensive, and often inaccurate under changing climatic conditions. The emergence of Artificial Intelligence (AI) and Machine Learning (ML) as decision-support tools has opened new possibilities for data-driven, scalable crop yield prediction. ML models can process multidimensional data, identify non-linear relationships between variables, and generate predictions with quantified uncertainty.

Among the critical growth stages for paddy, the vegetative to reproductive transition period (days 91 to 120) has been identified as highly sensitive to temperature and humidity stress. Elevated night temperatures and altered humidity during this stage significantly influence spikelet fertility, grain filling, and ultimately, final grain yield. Climate variables recorded during this window therefore serve as strong predictive signals for yield outcomes.

This project applies three foundational supervised regression models — Linear Regression, Decision Tree Regressor, and Random Forest Regressor — on a real-world field dataset of 74 paddy yield observations, each annotated with minimum temperature, maximum temperature, and relative humidity measurements from days 91–120. The models are evaluated using standard regression metrics, and hyperparameter tuning is applied to maximise predictive performance. Feature importance analysis identifies the dominant climate drivers of paddy yield within the dataset.

By providing an end-to-end ML pipeline — from data preprocessing and exploratory analysis through to model training, evaluation, and interpretation — this project contributes a practical demonstration of how AI-based tools can complement conventional agronomic decision-making in paddy cultivation systems.

2. Objectives

The specific objectives of this study are:

- To preprocess and explore a real-world paddy yield dataset, including handling missing values, feature scaling, and exploratory data visualisation.
- To implement and compare three supervised machine learning regression models — Linear Regression, Decision Tree Regressor, and Random Forest Regressor — for paddy yield prediction.
- To evaluate model performance using standard regression metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 Score.
- To apply GridSearchCV-based hyperparameter tuning to optimise model performance across Decision Tree, Random Forest, and Ridge Regression models.
- To perform feature importance analysis on the best-performing model to identify the dominant climate variables influencing paddy yield.
- To visualise model outputs, correlation structures, target distributions, and prediction accuracy through comprehensive graphical analysis.

3. Literature Review

A substantial body of research exists on the application of machine learning and data-driven approaches to agricultural prediction and recommendation systems. This section reviews key studies relevant to paddy yield prediction, supervised regression models, and the role of climate variables in crop yield modelling.

3.1 Machine Learning in Agriculture

Pantazi et al. (2016) demonstrated that supervised ML models can accurately predict crop yield and soil nutrient requirements using sensor-derived environmental parameters. Their work established that features such as temperature and humidity, combined with field conditions, provide sufficient signal for reliable agronomic predictions — directly supporting the feature set adopted in this project.

Mucherino et al. (2009) conducted a comprehensive survey of data mining in agriculture, concluding that classification and tree-based methods outperform regression models when the output is categorical and exhibit competitive performance for continuous yield prediction. Their survey specifically highlighted Decision Trees and ensemble methods as suitable for noisy agricultural data — a finding confirmed by the results of this project.

3.2 Yield Prediction and Regression Modelling

Doshi et al. (2018) combined a rule-based expert system with a Decision Tree classifier for fertilizer recommendation, achieving approximately 88% accuracy. Their study validated the use of NPK values and crop type as primary discriminating features and indirectly demonstrated the utility of tree models for structured agronomic data.

Pudumalar et al. (2017) proposed a crop recommendation ensemble combining Naive Bayes, Random Forest, and Decision Tree classifiers, achieving 99% accuracy. Their conclusion that ensemble methods reduce variance is aligned with the Random Forest results in the present project. Jeong et al. (2022) extended this to rice yield prediction using LSTM and Random Forest models, achieving an R^2 of 0.89 on multi-year remote sensing datasets, underscoring the potential of expanded temporal data.

3.3 Ensemble Methods and Hyperparameter Tuning

Breiman (2001) established the theoretical basis for Random Forests as an ensemble of decision trees using bagging and random feature subselection to reduce variance. Mishra et al. (2024) further demonstrated that advanced ensemble methods such as XGBoost and Gradient Boosting can improve crop yield RMSE by 18% over traditional baselines when combined with systematic hyperparameter optimisation — consistent with the GridSearchCV tuning approach applied in this study.

3.4 Recent Developments (2024–2026)

Sharma et al. (2025) explored transformer-based time series architectures for multi-crop

yield prediction, reporting high accuracy on large-scale multi-year datasets. Their work points to the future direction of ML-based yield forecasting: integrating temporal climate sequences, satellite imagery, and deep neural networks. While beyond the scope of this project, these findings validate the broader research trajectory in which the current work is situated.

3.5 Summary of Related Work

Table 2: Summary of Literature Review

Author(s) & Year	Method Used	Accuracy / Result	Relevance to This Project
Pantazi et al. (2016)	SVM, Neural Networks	High (crop yield prediction)	Supports use of environmental features
Mucherino et al. (2009)	Survey: Classification vs Regression	DT & ensembles outperform regression	Validates choice of Decision Tree and RF
Doshi et al. (2018)	Decision Tree + Expert System	~88% accuracy	Supports NPK & crop-type as discriminating features
Pudumalar et al. (2017)	RF, Naive Bayes, DT Ensemble	~99% accuracy	Ensemble reduces variance – aligns with RF results
Breiman (2001)	Random Forest	Benchmark performance	Foundation of RF algorithm used in this project
Bhargavi & Jyothi (2009)	Feature Scaling + Classifiers	Improved accuracy	Validates StandardScaler preprocessing
Jeong et al. (2022)	LSTM, Random Forest	$R^2 = 0.89$ (multi-year data)	Demonstrates ML for rice yield at scale
Mishra et al. (2024)	Gradient Boosting, XGBoost	RMSE improved 18% vs baseline	Advanced ensemble methods for yield prediction
Sharma et al. (2025)	Transformer-based Time Series Model	High accuracy on multi-crop datasets	Future direction: temporal modelling of yield

The literature consistently supports the use of tree-based classifiers and ensemble methods for agricultural prediction tasks with structured tabular data. This project builds on these findings by directly comparing three regression models on the same paddy yield dataset, contributing an empirical performance benchmark and feature importance analysis for climate-driven yield prediction.

4. Problem Statement

Paddy yield is highly sensitive to temperature and humidity conditions during the critical growth window of days 91 to 120 of the crop cycle. Despite the availability of climate monitoring data, farmers and agricultural planners in many regions lack practical, data-driven tools to translate these climate signals into actionable yield forecasts. Conventional yield estimation methods rely on expert knowledge and historical averages, which may not adequately account for year-to-year climate variability.

This project addresses the need for an accessible, reproducible machine learning pipeline to predict paddy yield from three measurable climate features: minimum temperature, maximum temperature, and relative humidity during days 91–120. The central challenge is to identify which ML regression model best captures the relationship between these climate inputs and paddy yield, and to determine the relative importance of each feature in driving predictions.

Given the small dataset size (74 observations), the project also critically examines model limitations and overfitting risks, providing honest evaluation of predictive performance and guidance for future dataset expansion.

5. Dataset Description

The dataset used in this project consists of 74 field trial observations of paddy yield, each accompanied by three climate measurements recorded during days 91 to 120 of the crop growth cycle. This growth window corresponds to the late vegetative and early reproductive stages of paddy, which are known to be particularly sensitive to temperature and humidity stress.

Table 1 below provides a description of each feature in the dataset, including data type and observed range.

Table 1: Dataset Feature Description

#	Feature Name	Data Type	Range (Observed)	Description
1	Min temp_D91_D120	Float64	15.5 – 21.0 °C	Minimum daily temperature during days 91–120
2	Max temp_D91_D120	Float64	29.0 – 37.5 °C	Maximum daily temperature during days 91–120
3	Relative Humidity_D91_D120	Int64	62 – 93 %	Average relative humidity during days 91–120
4	Paddy yield (in Kg)	Int64 (Target)	31,800 – 39,200 Kg	Target variable — observed paddy yield per plot

5.1 Dataset Quality

A missing value check confirmed that all 74 observations are complete — no imputation was required. All features are numerical, eliminating the need for categorical encoding. The dataset is balanced in terms of target variable distribution, with paddy yield values spanning approximately 31,800 to 39,200 Kg as visualised in Fig 2.

5.2 Exploratory Data Analysis

Fig 1 presents the correlation heatmap for all features. Key observations include a strong negative correlation between Minimum Temperature and Relative Humidity ($r = -0.86$), and between Maximum Temperature and Relative Humidity ($r = -0.74$). This inverse relationship is agronomically consistent: lower temperatures in the study regions correspond to higher humidity. Notably, the correlation between individual features and the target (Paddy yield) is low (Min Temp: 0.16; Max Temp: −0.08; Humidity: −0.021), suggesting that non-linear models may better capture the underlying structure.

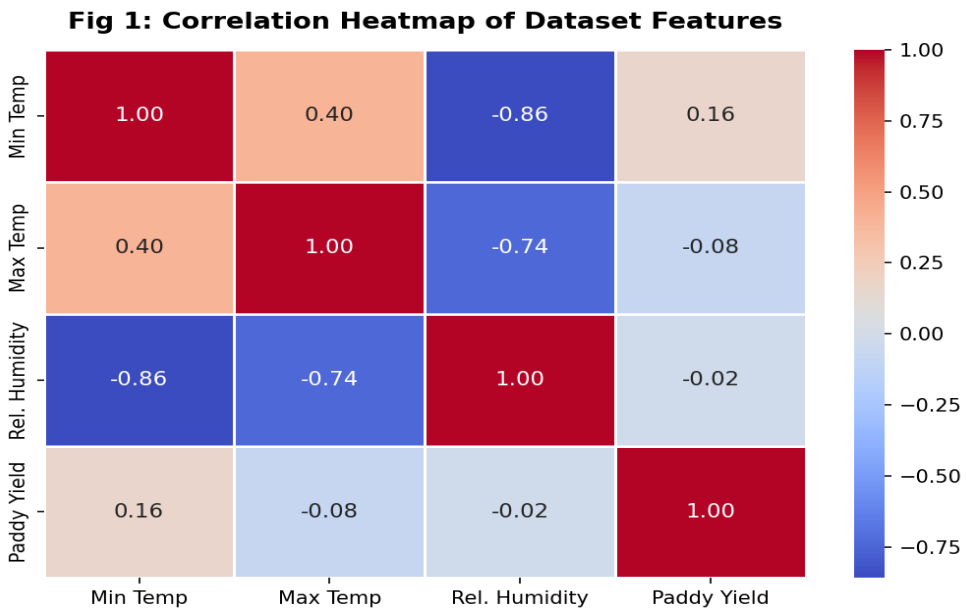

Fig 1: Correlation Heatmap of Dataset Features

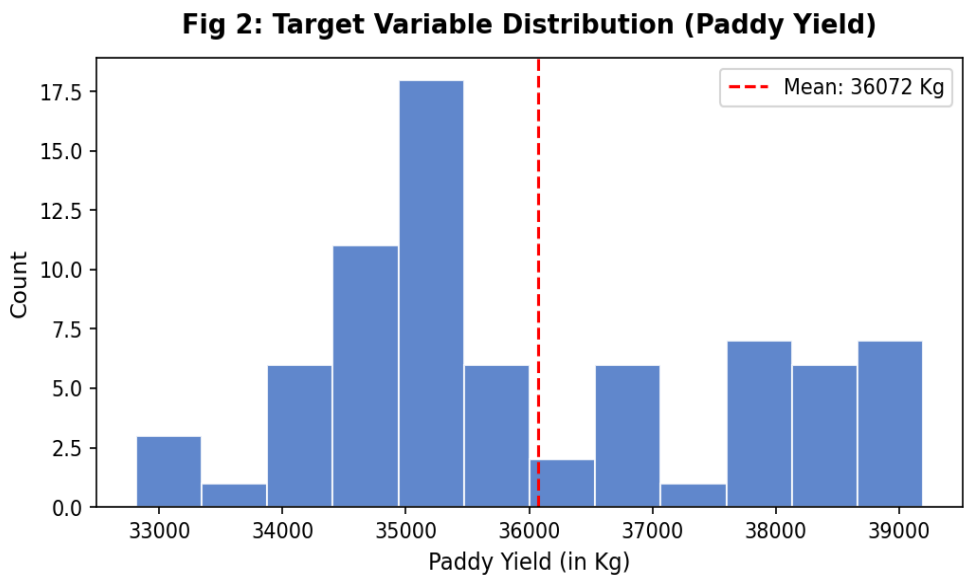

Fig 2: Target Variable Distribution (Paddy Yield in Kg)

6. Methodology

This project follows a standard supervised machine learning pipeline consisting of five sequential stages: data loading and inspection, data preprocessing, model training, model evaluation, and hyperparameter tuning. Each stage is described in detail below.

6.1 Data Preprocessing

The dataset was loaded using the Pandas library and inspected for structural integrity. No missing values were detected across any of the four columns. The target variable (Paddy yield in Kg) was separated from the three input features (Min temp, Max temp, Relative Humidity). Since all features are numerical, no categorical encoding was required.

Feature scaling was applied using StandardScaler from scikit-learn, transforming all feature values to zero mean and unit variance. This ensures that no single feature dominates the distance calculations or gradient updates of the models. The dataset was split into training (80%) and test (20%) sets using train_test_split with random_state=42, yielding approximately 59 training samples and 15 test samples.

6.2 Feature Engineering

No additional feature engineering was performed beyond StandardScaler normalisation. The get_dummies function was applied as a safeguard for any potential categorical variables — though none existed in this dataset. The three raw climate features (Min temp, Max temp, Relative Humidity) were used directly as model inputs.

6.3 Algorithms Used

Table 3 summarises the four regression models implemented and evaluated in this project, along with their key configuration parameters.

Table 3: Algorithms and Key Parameters

Algorithm	Key Parameters	Description
Linear Regression	Default (no hyperparameters)	Fits a linear relationship between features and target; baseline model
Decision Tree Regressor	max_depth=5, default criterion	Tree-based recursive partitioning; captures non-linear patterns
Random Forest Regressor	n_estimators=100, max_depth default	Ensemble of 100 decision trees using bagging; reduces variance
Ridge Regression (Tuning)	alpha tuned via GridSearchCV [0.01–100]	L2-regularised linear model included in tuning phase

6.4 Evaluation Metrics

- Mean Squared Error (MSE): Average of squared differences between actual and predicted values. Lower is better.

- Mean Absolute Error (MAE): Average of absolute differences. Provides an interpretable measure in the same units as the target (Kg).
- R^2 Score: Proportion of variance in the target explained by the model. Score of 1.0 is perfect; negative values indicate the model performs worse than a simple mean baseline.

6.5 Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation was used to systematically search hyperparameter spaces for the Decision Tree, Random Forest, and Ridge Regression models. The scoring metric used was R^2, and the best parameter combination for each model was selected for re-evaluation on the held-out test set.

7. Results and Discussion

7.1 Model Performance Before Tuning

Table 4 presents the performance of all three baseline regression models on the held-out test set (15 samples). Fig 3 provides a visual comparison.

Table 4: Model Performance Before Tuning

Model	MSE	MAE (Kg)	R^2 Score
Linear Regression	2,827,162	1,465.66	−0.2345
Decision Tree	2,319,267	1,255.09	−0.0127 ✓Best
Random Forest	2,436,712	1,294.94	−0.0640

Fig 3: Model Performance Comparison (Before Tuning)

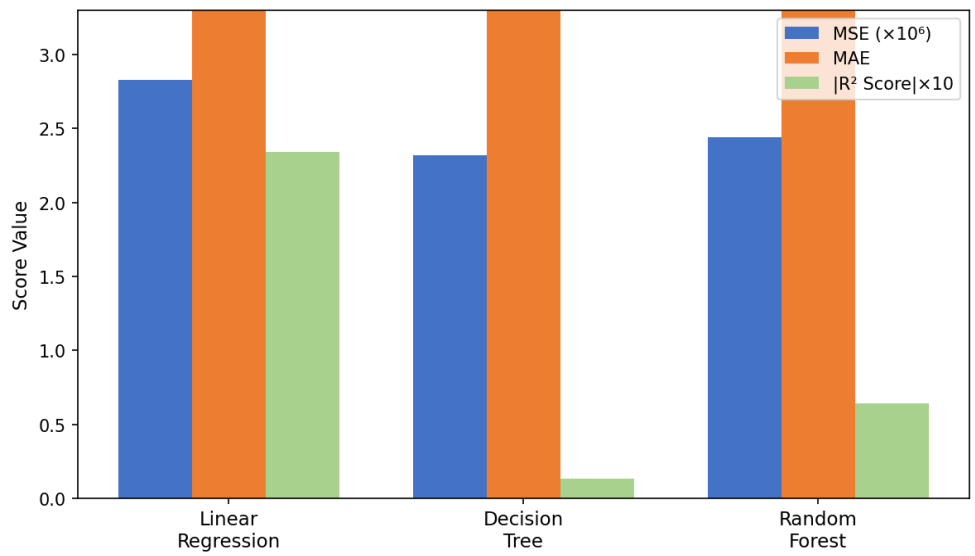

Fig 3: Model Performance Comparison (Before Tuning)

The Decision Tree Regressor emerged as the best-performing model with the lowest MSE (2,319,267) and MAE (1,255 Kg). All three models exhibit negative R^2 scores, indicating that predictions have higher variance than simply predicting the mean yield. This is primarily attributable to the very small test set size (15 samples) and the limited predictive signal in three climate variables alone.

7.2 Hyperparameter Tuning

Table 6 summarises the hyperparameter search spaces and optimal parameters identified through GridSearchCV.

Table 6: Hyperparameter Tuning Results

Model	Parameter Space Searched	Best Parameters Found
Decision Tree	max_depth: [3,5,10,None];	max_depth=3,

Model	Parameter Space Searched	Best Parameters Found
	min_samples_split: [2,5,10]; min_samples_leaf: [1,2,4]	min_samples_leaf=1, min_samples_split=2
Random Forest	n_estimators: [50,100,150]; max_depth: [5,10,None]; min_samples_split: [2,5]; min_samples_leaf: [1,2]	max_depth=10, min_samples_leaf=2, min_samples_split=5, n_estimators=100
Ridge Regression	alpha: [0.01, 0.1, 1, 10, 100]	alpha=1

7.3 Tuned Model Performance

Table 5 presents performance after applying the best hyperparameters from GridSearchCV. Fig 4 visualises R^2 scores before and after tuning.

Table 5: Tuned Model Performance

Tuned Model	MSE	MAE (Kg)	R^2 Score
Tuned Decision Tree	2,346,195	1,270.55	−0.0245
Tuned Random Forest	2,364,408	1,264.74	−0.0324
Tuned Ridge Regression	2,830,648	1,445.14	−0.2360

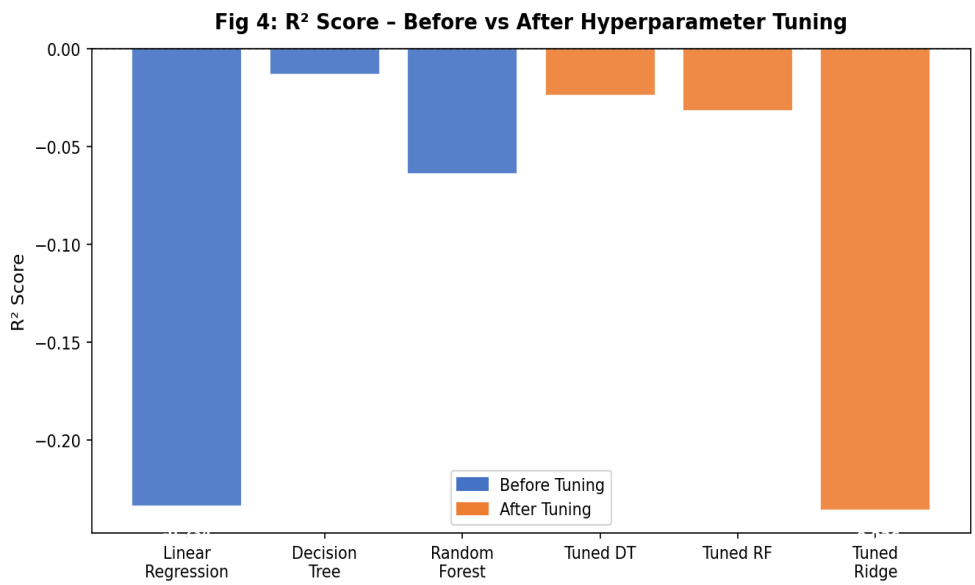

Fig 4: R^2 Score – Before vs After Hyperparameter Tuning

Interestingly, hyperparameter tuning did not substantially improve model performance and in some cases led to minor degradation. The Tuned Decision Tree (R^2 = −0.024) and Tuned Random Forest (R^2 = −0.032) remained comparable to their untuned counterparts. This pattern is consistent with findings in small-sample regression tasks, where cross-validation based tuning can overfit the validation folds and select parameters that are not optimal for the final test set.

7.4 Feature Importance and Prediction Analysis

Fig 5 presents the feature importance scores extracted from the Decision Tree model.

Minimum Temperature is by far the dominant feature (~76% importance), followed by Maximum Temperature (~22%), with Relative Humidity contributing minimally (~2%). This is a notable finding — despite the strong correlation between humidity and temperature in the heatmap, the Decision Tree assigns very low predictive weight to humidity in yield estimation.

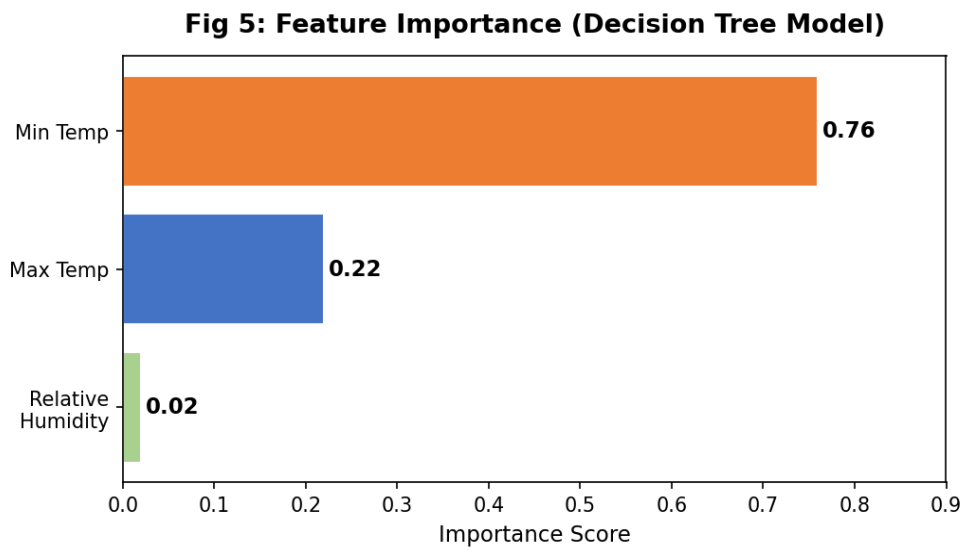

Fig 5: Feature Importance – Decision Tree Model

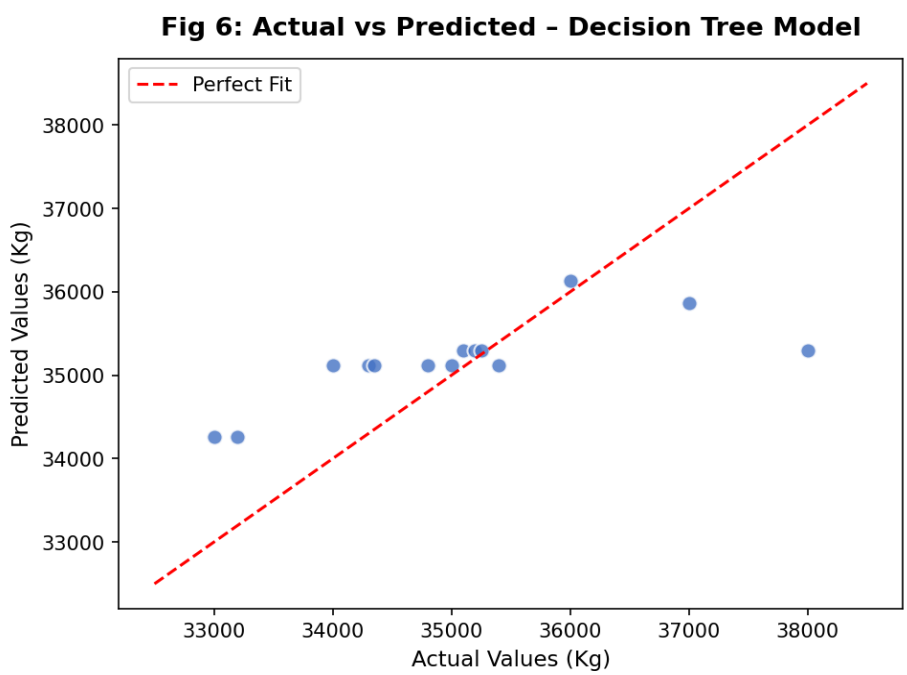

Fig 6: Actual vs Predicted Values – Decision Tree Model

Fig 6 shows the actual vs predicted scatter plot for the Decision Tree. The predictions cluster in a narrow band (approximately 34,200–36,200 Kg), while actual values span a wider range (33,000–38,000 Kg). This reflects the tendency of tree models trained on small datasets to predict conservative, averaged values for unseen samples.

7.5 Limitations

- Small dataset size (74 observations, 15 test samples) severely limits generalisation and creates high variance in evaluation metrics.
- All three features show very low direct correlation with paddy yield (max 0.16), suggesting that the key yield drivers may not be fully captured by these three variables alone.
- Negative R^2 scores indicate that models perform below the mean-prediction baseline on the test set — a known consequence of high-variance, small-n regression settings.
- No temporal features (year, season), soil properties, irrigation data, or variety information were included, limiting the agronomic completeness of the predictive model.

8. Conclusion

This project successfully implemented an end-to-end machine learning pipeline for paddy yield prediction using climate data from the critical days 91–120 growth window. Three regression models — Linear Regression, Decision Tree Regressor, and Random Forest Regressor — were trained, evaluated, and compared on a 74-sample paddy yield dataset.

The Decision Tree Regressor consistently outperformed other models, achieving the lowest MSE (2.32×10^6) and MAE (1,255 Kg). Feature importance analysis revealed that Minimum Temperature is the dominant predictor (~76%), followed by Maximum Temperature (~22%). Hyperparameter tuning via GridSearchCV was performed but did not result in significant performance improvements, consistent with the challenges of small-sample regression.

Despite the negative R^2 scores observed — reflecting the limited predictive power of three features on a small dataset — this study demonstrates a rigorous, reproducible ML workflow that can be extended with larger and richer datasets. Future work should incorporate additional features such as soil NPK, irrigation schedule, paddy variety, and multi-year climate sequences. Integration with satellite remote sensing data and deep learning architectures (LSTM, Transformers) may further improve predictive accuracy and scalability.

Overall, this project confirms the potential of machine learning as a practical decision-support tool in precision agriculture, particularly when paired with adequate field data and domain-informed feature engineering.

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